



SWITCH Cloud Portal S3:

A look into the future and its current status

Theofilos Mouratidis, Ceph Science User Group

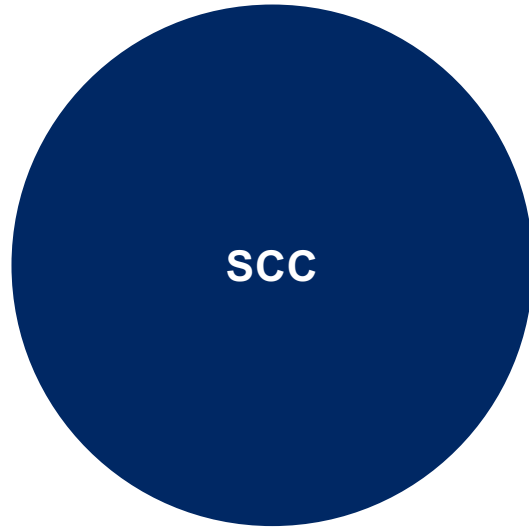
Version 1.0, 21 May 2026

Who are we?

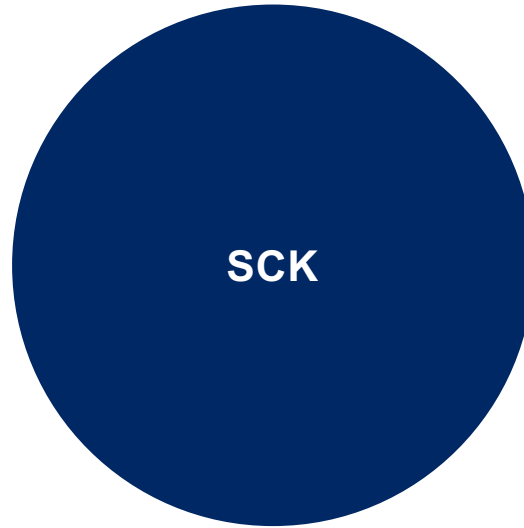


- Switch is the NREN of Switzerland:
 - edu-id
 - Swiss e-ID
 - Authority for “.ch” domains
 - Switch Cloud Portal
 - And many more other projects
- We work in the YODA circle of the Switch Cloud team
 - YOur DAta(centers)
- We specialise in DC management and Storage solutions
 - Ceph (S3)
 - IBM Flashstorage (Openstack Cinder)
 - NetApp (Openstack Cinder/Manila)

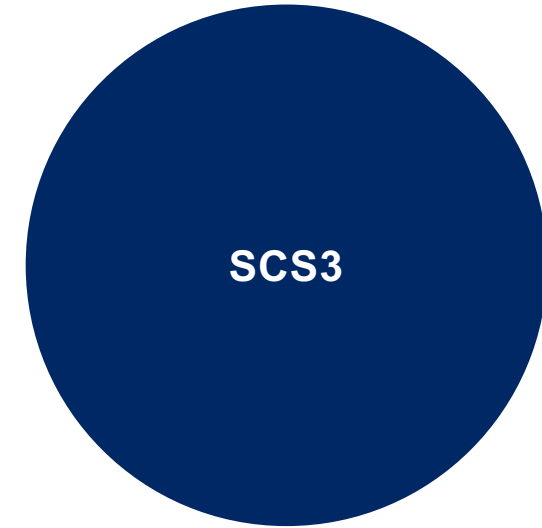
Switch Cloud Portal – A broker for various services



Switch Cloud Compute:
Openstack based VM service



Switch Cloud Kubernetes:
KaaS based on SCC



Switch Cloud S3: Self-service S3 projects with light management

Ceph in YODA

YODA uses Ceph to provide the S3 service for Switch Cloud portal

- We have 3 prod clusters (in 2 DCs), 28 PiB usage storage
- We run Ceph v19.2.3 + vulnerability backports + custom patches
 - We use Ubuntu 22.04 for prod and 24.04 for testing and stage environments
 - Our patches include our improvements that have been merged to main but not backported yet
 - Mainly minor bugs or cephadm/ceph orch features/improvements
- Our biggest use-case is S3
 - There were plans for Manila and multi-attach volumes for K8s but NetApp was chosen instead
- We use Terraform and Ansible to deploy and configure our clusters
- We use Gitlab pipelines to push configuration to the Ceph servers
- We have test/staging/prod environments
 - Prod: Physical clusters (integrate with Switch Cloud Portal)
 - Staging: Long lasting VM clusters (integrate with Switch Cloud Portal)
 - Test: VM clusters that last within a Gitlab pipeline

Switch Cloud Portal S3

SCS3 is one of the services the SCP provides, mainly focused on archival and backups.

- SCP is a service broker
- Each service is provided via a middleware
 - They are called service adapters
- In our case, the Portal team developed the S3 SA
 - Ruby on Rails backend
 - Interaction with Cephadm API
 - Automatic workflows for user management
 - We have minimal interaction with radosgw-admin
 - We can focus more on keeping the Ceph service highly available

The screenshot shows the Switch Cloud Portal interface for user Theofilos Mouratidis. The top navigation bar includes the Switch Cloud logo and links for 'What's New?', 'Documentation', and 'Help Desk'. The left sidebar contains icons for navigation. The main content area is titled 'Theofilos Mouratidis #OTUm' and displays four service cards: 'Compute', 'Kubernetes', 'S3', and 'Project Members'. Each card shows a status (Active), a set of icons representing different resources, and a 'See more details' link.

Service	Status	Resources	Details Link
Compute	Active	2 Users, 0 Keys, 19 CPUs, 40 Disks, 0 Cameras, 6 Groups, 5 Globes, 5 Antennas, 4 Networks	See more details
Kubernetes	Active	1 Cluster, 1 User	See more details
S3	Active	0 Buckets, 3 Keys, 0 Files, 0 Bytes	See more details
Project Members		1 Member	Manage members

SCS3 in detail

SwitchCloud

What's New? Documentation Help Desk

Theofilos Mouratidis

← Back to organisation

Theofilos Mouratidis

Compute

Kubernetes

S3

Billing

Members

Settings

CURE > Theofilos Mouratidis > S3

S3

Authorization Bucket Management Quota Details

Access Keys

Create access key +

Access keys are used to authenticate applications with the S3 API. After creation, the secret key will be shown only once.

Name	Access key id	Endpoint	Created	Actions
tsoko Active	...a759470	ZHC-C (HDD)	7 months ago	Edit Delete
key-0194dbdd32427909bf87a5b001b98ad1 Active	...1b98ad1	ZHW-B (HDD)	12 months ago	Edit Delete
key-019561d0b7c67fd0ada258b7b51e97f0 Active	...51e97f0	ZHW-A (SSD)	12 months ago	Edit Delete

Moving forward with SCS3

Switch Cloud Portal S3 now vs the future

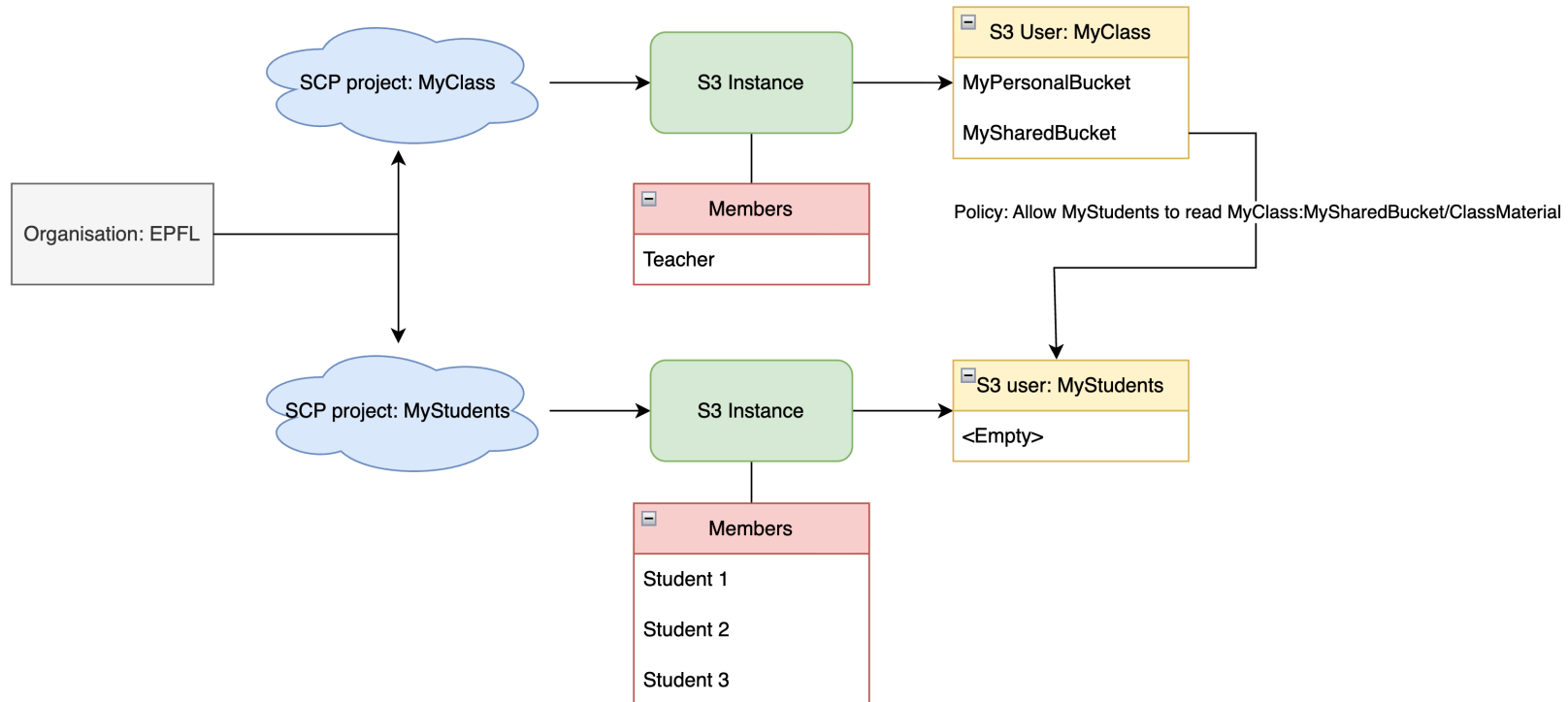
The current service adapter uses a combination of calls to the radosgw-admin and S3 APIs

- Single user per project
- No way to distinguish project members
- RBAC becomes unmanageable
 - Multiple projects for access control
- SA acts as the user to create buckets
 - Security issue?

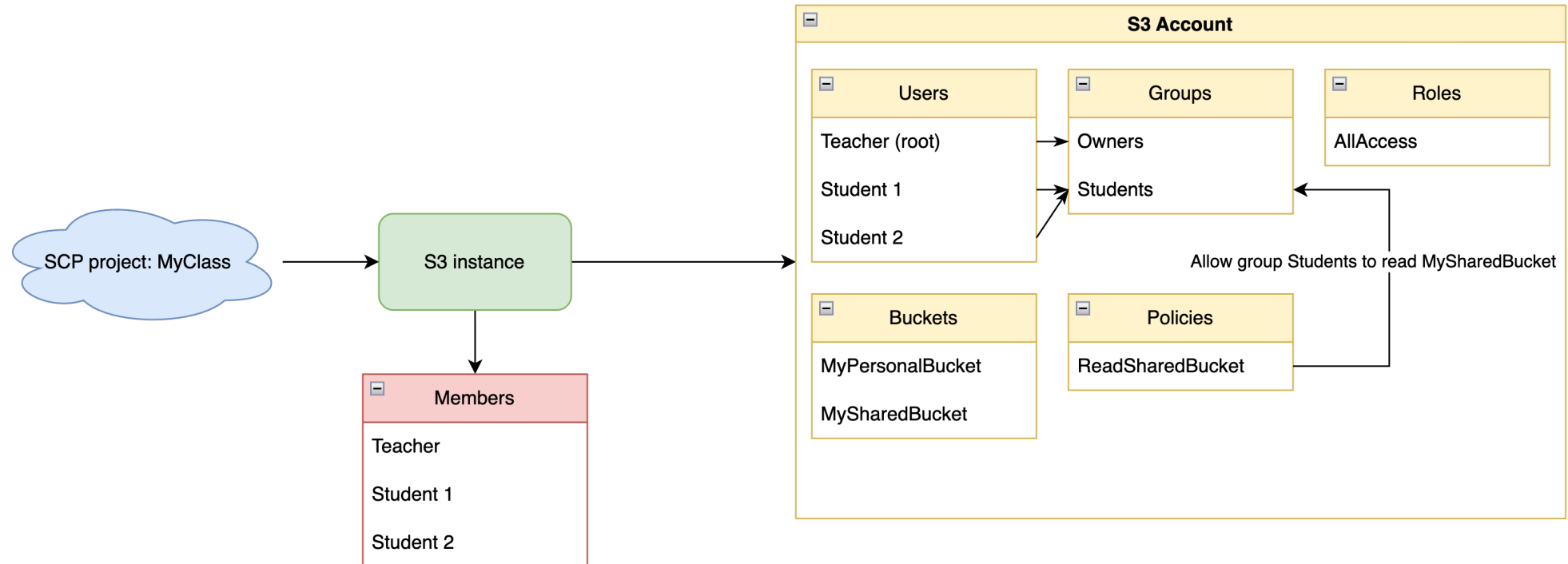
Moving into accounts and using the AWS IAM API for access control.

- RGW Account per project
- Better RBAC Management + STS
- Split responsibilities of the SA
 - Radosgw-api calls remain in the portal
 - S3 calls move to a dedicated dashboard
 - Like Horizon for Openstack
- The new Dashboard should manage
 - Users/members
 - Buckets + Bucket browser
 - Quota
 - Access keys
 - Policies (bucket/iam)

Example: Current SCS3



Example: Next step for SCS3 (in progress)



Questions?

YODA-Storage



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